

PU5558-PU5567-computer-lab-report

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Introduction

The following report is based on Provisional Patient Reported Outcome Measures (PROMS) specifically on the outcome of Hip and Knee Replacement procedures completed between April 2020 to March 2021, provided by NHS England. For this report the focus will be on the outcome measures of knee replacement surgery. Specific outcome measures pertaining to the Quality Of Life (QOL) to the patient have been recorded through a visual analogue scale designed to measure the self-reported status of the patients quality of life pre/post surgery. These values gathered from the survey will be used to inform the models.

Load necessary packages

```
library(tidyverse)
library(tidymodels)
library(corrplot)
library(psych)
library(ggplot2)
library(rmarkdown)
library(reshape2)
library(vip)
library(Hmisc)
library(VIM)
library(caret)
library(flextable)
library(gtsummary)
```

Load chosen dataset

```
kneeData <- read_csv("Knee_Replacement_CCG 2021.csv")
kneeData <- kneeData %>%
  select(-Procedure, -Year, -"Revision Flag", -"Provider Code") %>%
  drop_na() %>%
  unique()
```

```
glimpse(kneeData)
```

```
## Rows: 4,528
## Columns: 77
## $ `Age Band` <chr> "*", "*", "*", "*",~
## $ Gender <chr> "*", "*", "*", "*",~
## $ `Pre-Op Q Assisted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Assisted By` <dbl> 0, 0, 0, 0, 0, 0, 0~
## $ `Pre-Op Q Symptom Period` <dbl> 2, 4, 4, 2, 4, 2, 2~
## $ `Pre-Op Q Previous Surgery` <dbl> 2, 1, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Disability` <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Heart Disease` <dbl> 9, 9, 9, 9, 9, 1, 9~
## $ `High Bp` <dbl> 1, 9, 9, 1, 1, 1, 9~
## $ Stroke <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Circulation <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Lung Disease` <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ Diabetes <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ `Kidney Disease` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Nervous System` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Liver Disease` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Cancer <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Depression <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Arthritis <dbl> 9, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Mobility` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Self-Care` <dbl> 1, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Activity` <dbl> 3, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Discomfort` <dbl> 2, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Anxiety` <dbl> 1, 1, 2, 2, 1, 2, 1~
## $ `Pre-Op Q EQ5D Index Profile` <dbl> 21321, 21221, 21332~
## $ `Pre-Op Q EQ5D Index` <dbl> 0.364, 0.691, 0.030~
## $ `Post-Op Q Assisted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Assisted By` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Post-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Post-Op Q Disability` <dbl> 1, 2, 2, 2, 2, 1, 2~
## $ `Post-Op Q Mobility` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Self-Care` <dbl> 2, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Activity` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Discomfort` <dbl> 1, 1, 1, 1, 1, 2, 2~
## $ `Post-Op Q Anxiety` <dbl> 2, 1, 1, 1, 1, 1, 1~
## $ `Post-Op Q Satisfaction` <dbl> 2, 3, 1, 2, 1, 4, 3~
## $ `Post-Op Q Sucess` <dbl> 1, 1, 1, 1, 1, 4, 1~
## $ `Post-Op Q Allergy` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Bleeding` <dbl> 2, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Wound` <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Urine` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Further Surgery` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Readmitted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q EQ5D Index Profile` <dbl> 12112, 11111, 11111~
## $ `Post-Op Q EQ5D Index` <dbl> 0.744, 1.000, 1.000~
## $ `Knee Replacement EQ 5D Index Post-Op Q Predicted` <dbl> 0.6621424, 0.740953~
## $ `Pre-Op Q EQ VAS` <dbl> 85, 50, 46, 60, 60,~
## $ `Post-Op Q EQ VAS` <dbl> 60, 100, 90, 95, 90~
## $ `Knee Replacement EQ VAS_Post-Op Q Predicted` <dbl> 75.63073, 66.38606,~
```

```
## $ `Knee Replacement Pre-Op Q Pain` <dbl> 1, 1, 0, 0, 1, 1, 1~
## $ `Knee Replacement Pre-Op Q Night Pain` <dbl> 1, 2, 2, 2, 2, 3, 0~
## $ `Knee Replacement Pre-Op Q Washing` <dbl> 3, 4, 3, 4, 3, 3, 3~
## $ `Knee Replacement Pre-Op Q Transport` <dbl> 1, 4, 2, 4, 2, 2, 2~
## $ `Knee Replacement Pre-Op Q Walking` <dbl> 3, 3, 2, 3, 3, 1, 2~
## $ `Knee Replacement Pre-Op Q Standing` <dbl> 3, 2, 2, 3, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Limping` <dbl> 0, 3, 0, 0, 3, 1, 1~
## $ `Knee Replacement Pre-Op Q Kneeling` <dbl> 0, 0, 0, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Work` <dbl> 2, 2, 1, 3, 2, 1, 2~
## $ `Knee Replacement Pre-Op Q Confidence` <dbl> 3, 3, 3, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Shopping` <dbl> 2, 4, 0, 2, 3, 0, 2~
## $ `Knee Replacement Pre-Op Q Stairs` <dbl> 1, 4, 2, 2, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Score` <dbl> 20, 32, 17, 27, 27, ~
## $ `Knee Replacement Post-Op Q Pain` <dbl> 4, 3, 3, 4, 3, 1, 2~
## $ `Knee Replacement Post-Op Q Night Pain` <dbl> 4, 4, 3, 4, 4, 3, 2~
## $ `Knee Replacement Post-Op Q Washing` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Transport` <dbl> 2, 4, 4, 4, 4, 3, 4~
## $ `Knee Replacement Post-Op Q Walking` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Standing` <dbl> 3, 4, 4, 4, 4, 1, 3~
## $ `Knee Replacement Post-Op Q Limping` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Kneeling` <dbl> 0, 4, 0, 3, 2, 0, 2~
## $ `Knee Replacement Post-Op Q Work` <dbl> 3, 3, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Confidence` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Shopping` <dbl> 4, 4, 4, 4, 4, 0, 4~
## $ `Knee Replacement Post-Op Q Stairs` <dbl> 3, 4, 4, 4, 3, 1, 4~
## $ `Knee Replacement Post-Op Q Score` <dbl> 37, 46, 42, 47, 44, ~
## $ `Knee Replacement OKS Post-Op Q Predicted` <dbl> 34.02619, 36.88715, ~
```

```
# glimpseData <- kneeData[, c("Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
# "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
# "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index")]
# kable(glimpseData, format = "latex", booktabs = TRUE, longtable = TRUE,
# caption="Rate of Change within an Individual Health Board Against its Moving Average") %>%
# kable_styling(latex_options = c("striped", "hold_position")) %>%
# row_spec(0, bold = TRUE) %>%
# column_spec(1, "2cm") %>%
# row_spec(15, bold = TRUE) %>%
# kableExtra::landscape()

# set_flextable_defaults(
# font.family = "Arial", font.size = 10,
# border.color = "gray", big.mark = ""
# )

# ft <- flextable(head(kneeData), col_keys = c(
# "Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
# "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
# "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index"
# )) |> bold(part = "header")
# set_table_properties(ft, width=.5, layout = "autofit")
```

Characteristic	N = 4,528 ¹
Age Band	
*	1,603 (35%)
50 to 59	164 (3.6%)
60 to 69	1,034 (23%)
70 to 79	1,511 (33%)
80 to 89	216 (4.8%)
Gender	
*	1,603 (35%)
1	1,326 (29%)
2	1,599 (35%)
Knee Replacement Post-Op Q Pain	
0	116 (2.6%)
1	773 (17%)
2	974 (22%)
3	1,529 (34%)
4	1,136 (25%)
Post-Op Q EQ5D Index	0.76 (0.69, 1.00)
Pre-Op Q Previous Surgery	
1	437 (9.7%)
2	4,060 (90%)
9	31 (0.7%)

¹n (%); Median (Q1, Q3)

Dataset description

This dataset uses data from NHS England regarding PROMS for knee replacement surgery. The way in which this data has been gathered is to collect self-reported symptoms/QOL metrics through two standard procedures. One survey is the EQ-5DTM Index designed to evaluate the QOL in terms of of doing daily tasks such as(cite NHS PROMS): * Ability to pursue usual activites * Current experience of anxiety and/or depression, if any * Current experience of pain and discomfort if any * Mobility * Ability to wash and dress themselves

While the EQ5 focuses on day to day tasks the EQ-VAS focuses on pre/post operative outcomes through a visual thermomemter scale rannging from 0 to 100. This scale is designed to aid patients in describing how they feel post operation about the results and if their problems pre operation have been addressed with the surgery.

Visually inspecting the database we see that Age Band and Gender have many missing values. Alongside this, there are many missing values for co-morbidites. As instructed all missing values for co-morbidites denotes as 9 will be taken as a no value. However, since Age Band and Gender typically are important factors (especially as confounders), these values will be imputed. Since there are a possible 77 variables to provide summary statistics on, only specific columns relevant to the following questions below will be considered, stratified by pre/post surgery.

```
kneeData |> tbl_summary(include=c("Age Band", "Gender", "Knee Replacement Post-Op Q Pain", "Post-Op Q EQ5D
# describe(kneeData, fast=TRUE)
```

The following table provides some basic summary statistics on what the data represents ## Suitable machine learning algorithm for three questions:

1. Before the operation, can we estimate the post-operative EQ5D index for a patient? Given certain presence of certain predictors being present, this would fall into supervised learning there are no missing values so no semi supervised is needed 0.364, 0.691, can use simple linear regression then test how well the model performs with different AIC values. This question will be used for this report
2. Before the operation, can we predict how much pain a patient will have after the operation? Since pain is measured in categories we would want a classifier to predict which index the post operative pain will be in this case we would use an unsupervised algorithm, seems intuitive that you would use forests

0 = Severe 1 = Moderate 2 = Mild 3 = Very Mild 4 = None 9 = Missing

3. Before the operation, can we calculate how many patients have had previous surgery? Here we can use supervised/semi-supervised learning methods to calculate the numerical value of the patients that have had the previous surgery. The results is binary whether they have or have not had it

Model building to answer chosen question

1. Data splitting

```
kneeData["Age Band"][kneeData["Age Band"] == "50 to 59"] <- "1"
kneeData["Age Band"][kneeData["Age Band"] == "60 to 69"] <- "2"
kneeData["Age Band"][kneeData["Age Band"] == "70 to 79"] <- "3"
kneeData["Age Band"][kneeData["Age Band"] == "80 to 89"] <- "4"
kneeData$"Age Band" <- as.numeric(as.character(kneeData$"Age Band"))
kneeData["Age Band"][is.na(kneeData["Age Band"])] <- median(kneeData$"Age Band", na.rm = TRUE)
kneeData[kneeData=="*"] <- NA
imputedGender <- kNN(kneeData, k = 10)
kneeData$Gender <- imputedGender$Gender
kneeData$Gender <- as.numeric(kneeData$Gender)

kneeSplit <- kneeData %>%
  initial_split(prop = 0.7, strata = "Post-Op Q EQ5D Index")
kneeTrain <- training(kneeSplit)
kneeTest <- testing(kneeSplit)
```

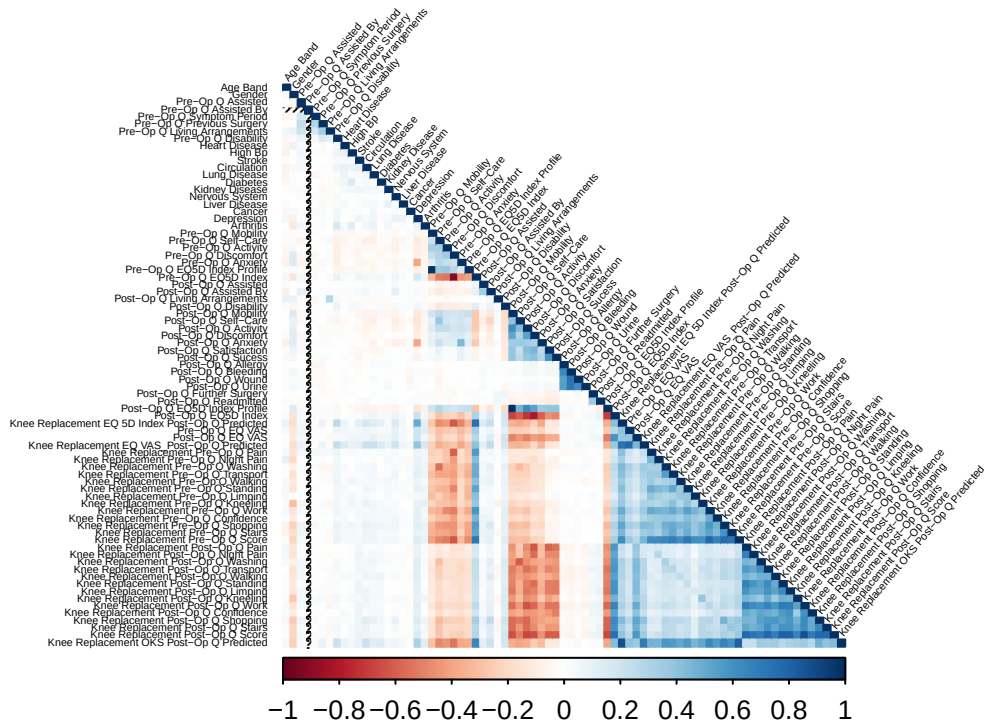
2. Selection and preprocessing of predictors A quick analysis of the data shows that there are several variables that have a high proportion of missing values. This is represented with the median value being the missing value "9" this level of missing data will have severe effects on the true relationship between the post-op pain and the independent variables. Deleting these values would not be suitable as they contain relevant information that could have affect post-op pain (reference how co-morbidities increase complications/pain management post-op). As to the specifications of this assignment these values will be recoded so the value "9" represents "No".

Categorical values in the Age Band need to be recoded into dummy variables for regression analysis, the following encodings have been used in replacement of the original categories

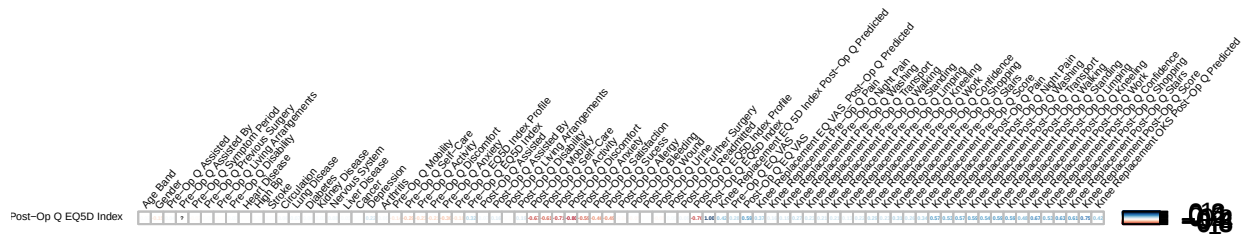
1- 50 to 59 2- 60-69 3- 70-79 4- 80-89

Both Gender and Age Band have a considerable percentage of missing values equating to almost half the dataset, these will be addressed with different methods. Age Band will use imputation in the form of imputating the median of the categories from the encodings (1-4). Gender will use KNN imputation to maintain some randomness to the distribution.

```
kneeCorr <- cor(kneeTrain)
corrplot(kneeCorr, method = "color", type = "lower", tl.col = "black", tl.srt = 50, number.cex = 0.5, t
```



```
kneeCorrPostOp <- t(kneeCorr[, c(46), drop = FALSE])
corrplot(kneeCorrPostOp, method = "number", tl.col = "black", tl.srt = 50, tl.cex = 0.3, cex.axis=0.3, num
```



```
for (i in seq_along(kneeTrain)) {
  cat("Index:", i, "- Name:", colnames(kneeTrain)[i], "\n")
}
```

```
## Index: 1 - Name: Age Band
## Index: 2 - Name: Gender
## Index: 3 - Name: Pre-Op Q Assisted
## Index: 4 - Name: Pre-Op Q Assisted By
## Index: 5 - Name: Pre-Op Q Symptom Period
## Index: 6 - Name: Pre-Op Q Previous Surgery
## Index: 7 - Name: Pre-Op Q Living Arrangements
## Index: 8 - Name: Pre-Op Q Disability
## Index: 9 - Name: Heart Disease
## Index: 10 - Name: High Bp
## Index: 11 - Name: Stroke
## Index: 12 - Name: Circulation
## Index: 13 - Name: Lung Disease
## Index: 14 - Name: Diabetes
## Index: 15 - Name: Kidney Disease
## Index: 16 - Name: Nervous System
## Index: 17 - Name: Liver Disease
## Index: 18 - Name: Cancer
## Index: 19 - Name: Depression
## Index: 20 - Name: Arthritis
## Index: 21 - Name: Pre-Op Q Mobility
## Index: 22 - Name: Pre-Op Q Self-Care
```

Index: 23 - Name: Pre-Op Q Activity
 ## Index: 24 - Name: Pre-Op Q Discomfort
 ## Index: 25 - Name: Pre-Op Q Anxiety
 ## Index: 26 - Name: Pre-Op Q EQ5D Index Profile
 ## Index: 27 - Name: Pre-Op Q EQ5D Index
 ## Index: 28 - Name: Post-Op Q Assisted
 ## Index: 29 - Name: Post-Op Q Assisted By
 ## Index: 30 - Name: Post-Op Q Living Arrangements
 ## Index: 31 - Name: Post-Op Q Disability
 ## Index: 32 - Name: Post-Op Q Mobility
 ## Index: 33 - Name: Post-Op Q Self-Care
 ## Index: 34 - Name: Post-Op Q Activity
 ## Index: 35 - Name: Post-Op Q Discomfort
 ## Index: 36 - Name: Post-Op Q Anxiety
 ## Index: 37 - Name: Post-Op Q Satisfaction
 ## Index: 38 - Name: Post-Op Q Success
 ## Index: 39 - Name: Post-Op Q Allergy
 ## Index: 40 - Name: Post-Op Q Bleeding
 ## Index: 41 - Name: Post-Op Q Wound
 ## Index: 42 - Name: Post-Op Q Urine
 ## Index: 43 - Name: Post-Op Q Further Surgery
 ## Index: 44 - Name: Post-Op Q Readmitted
 ## Index: 45 - Name: Post-Op Q EQ5D Index Profile
 ## Index: 46 - Name: Post-Op Q EQ5D Index
 ## Index: 47 - Name: Knee Replacement EQ 5D Index Post-Op Q Predicted
 ## Index: 48 - Name: Pre-Op Q EQ VAS
 ## Index: 49 - Name: Post-Op Q EQ VAS
 ## Index: 50 - Name: Knee Replacement EQ VAS_Post-Op Q Predicted
 ## Index: 51 - Name: Knee Replacement Pre-Op Q Pain
 ## Index: 52 - Name: Knee Replacement Pre-Op Q Night Pain
 ## Index: 53 - Name: Knee Replacement Pre-Op Q Washing
 ## Index: 54 - Name: Knee Replacement Pre-Op Q Transport
 ## Index: 55 - Name: Knee Replacement Pre-Op Q Walking
 ## Index: 56 - Name: Knee Replacement Pre-Op Q Standing
 ## Index: 57 - Name: Knee Replacement Pre-Op Q Limping
 ## Index: 58 - Name: Knee Replacement Pre-Op Q Kneeling
 ## Index: 59 - Name: Knee Replacement Pre-Op Q Work
 ## Index: 60 - Name: Knee Replacement Pre-Op Q Confidence
 ## Index: 61 - Name: Knee Replacement Pre-Op Q Shopping
 ## Index: 62 - Name: Knee Replacement Pre-Op Q Stairs
 ## Index: 63 - Name: Knee Replacement Pre-Op Q Score
 ## Index: 64 - Name: Knee Replacement Post-Op Q Pain
 ## Index: 65 - Name: Knee Replacement Post-Op Q Night Pain
 ## Index: 66 - Name: Knee Replacement Post-Op Q Washing
 ## Index: 67 - Name: Knee Replacement Post-Op Q Transport
 ## Index: 68 - Name: Knee Replacement Post-Op Q Walking
 ## Index: 69 - Name: Knee Replacement Post-Op Q Standing
 ## Index: 70 - Name: Knee Replacement Post-Op Q Limping
 ## Index: 71 - Name: Knee Replacement Post-Op Q Kneeling
 ## Index: 72 - Name: Knee Replacement Post-Op Q Work
 ## Index: 73 - Name: Knee Replacement Post-Op Q Confidence
 ## Index: 74 - Name: Knee Replacement Post-Op Q Shopping
 ## Index: 75 - Name: Knee Replacement Post-Op Q Stairs
 ## Index: 76 - Name: Knee Replacement Post-Op Q Score


```
## Index: 77 - Name: Knee Replacement OKS Post-Op Q Predicted
```

The full correlation plot is displayed above but the values and their labels are difficult to read with the 77 variables that we have to consider. Extrcation of significant correlations in the main plot will be used to hilight the variables that will be used as predictors for the model. Correlation values were observed through the plots and all predictor values that were above 0.5 or below -0.5 were considered as significant predictors. This is puerly numerically based, but given clinical expertise there may be some predictors that may want to be included into the final model.

```
colnames(kneeTrain) <- make.names(colnames(kneeTrain))
colnames(kneeTest) <- make.names(colnames(kneeTest))
print(colnames(kneeTrain))
```

```
## [1] "Age.Band"
## [2] "Gender"
## [3] "Pre.Op.Q.Assisted"
## [4] "Pre.Op.Q.Assisted.By"
## [5] "Pre.Op.Q.Symptom.Period"
## [6] "Pre.Op.Q.Previous.Surgery"
## [7] "Pre.Op.Q.Living.Arrangements"
## [8] "Pre.Op.Q.Disability"
## [9] "Heart.Disease"
## [10] "High.Bp"
## [11] "Stroke"
## [12] "Circulation"
## [13] "Lung.Disease"
## [14] "Diabetes"
## [15] "Kidney.Disease"
## [16] "Nervous.System"
## [17] "Liver.Disease"
## [18] "Cancer"
## [19] "Depression"
## [20] "Arthritis"
## [21] "Pre.Op.Q.Mobility"
## [22] "Pre.Op.Q.Self.Care"
## [23] "Pre.Op.Q.Activity"
## [24] "Pre.Op.Q.Discomfort"
## [25] "Pre.Op.Q.Anxiety"
## [26] "Pre.Op.Q.EQ5D.Index.Profile"
## [27] "Pre.Op.Q.EQ5D.Index"
## [28] "Post.Op.Q.Assisted"
## [29] "Post.Op.Q.Assisted.By"
## [30] "Post.Op.Q.Living.Arrangements"
## [31] "Post.Op.Q.Disability"
## [32] "Post.Op.Q.Mobility"
## [33] "Post.Op.Q.Self.Care"
## [34] "Post.Op.Q.Activity"
## [35] "Post.Op.Q.Discomfort"
## [36] "Post.Op.Q.Anxiety"
## [37] "Post.Op.Q.Satisfaction"
## [38] "Post.Op.Q.Sucess"
## [39] "Post.Op.Q.Allergy"
## [40] "Post.Op.Q.Bleeding"
## [41] "Post.Op.Q.Wound"
## [42] "Post.Op.Q.Urine"
```

```

## [43] "Post.Op.Q.Further.Surgery"
## [44] "Post.Op.Q.Readmitted"
## [45] "Post.Op.Q.EQ5D.Index.Profile"
## [46] "Post.Op.Q.EQ5D.Index"
## [47] "Knee.Replacement.EQ.5D.Index.Post.Op.Q.Predicted"
## [48] "Pre.Op.Q.EQ.VAS"
## [49] "Post.Op.Q.EQ.VAS"
## [50] "Knee.Replacement.EQ.VAS_Post.Op.Q.Predicted"
## [51] "Knee.Replacement.Pre.Op.Q.Pain"
## [52] "Knee.Replacement.Pre.Op.Q.Night.Pain"
## [53] "Knee.Replacement.Pre.Op.Q.Washing"
## [54] "Knee.Replacement.Pre.Op.Q.Transport"
## [55] "Knee.Replacement.Pre.Op.Q.Walking"
## [56] "Knee.Replacement.Pre.Op.Q.Standing"
## [57] "Knee.Replacement.Pre.Op.Q.Limping"
## [58] "Knee.Replacement.Pre.Op.Q.Kneeling"
## [59] "Knee.Replacement.Pre.Op.Q.Work"
## [60] "Knee.Replacement.Pre.Op.Q.Confidence"
## [61] "Knee.Replacement.Pre.Op.Q.Shopping"
## [62] "Knee.Replacement.Pre.Op.Q.Stairs"
## [63] "Knee.Replacement.Pre.Op.Q.Score"
## [64] "Knee.Replacement.Post.Op.Q.Pain"
## [65] "Knee.Replacement.Post.Op.Q.Night.Pain"
## [66] "Knee.Replacement.Post.Op.Q.Washing"
## [67] "Knee.Replacement.Post.Op.Q.Transport"
## [68] "Knee.Replacement.Post.Op.Q.Walking"
## [69] "Knee.Replacement.Post.Op.Q.Standing"
## [70] "Knee.Replacement.Post.Op.Q.Limping"
## [71] "Knee.Replacement.Post.Op.Q.Kneeling"
## [72] "Knee.Replacement.Post.Op.Q.Work"
## [73] "Knee.Replacement.Post.Op.Q.Confidence"
## [74] "Knee.Replacement.Post.Op.Q.Shopping"
## [75] "Knee.Replacement.Post.Op.Q.Stairs"
## [76] "Knee.Replacement.Post.Op.Q.Score"
## [77] "Knee.Replacement.OKS.Post.Op.Q.Predicted"

```

```

kneeModelRec <- kneeTrain %>%
  recipe(Post.Op.Q.EQ5D.Index ~
    Post.Op.Q.Mobility +
    Post.Op.Q.Self.Care +
    Post.Op.Q.Discomfort +
    Post.Op.Q.EQ5D.Index.Profile +
    Post.Op.Q.EQ.VAS +
    Knee.Replacement.Post.Op.Q.Pain +
    Knee.Replacement.Post.Op.Q.Night.Pain +
    Knee.Replacement.Post.Op.Q.Washing +
    Knee.Replacement.Post.Op.Q.Transport +
    Knee.Replacement.Post.Op.Q.Walking +
    Knee.Replacement.Post.Op.Q.Standing +
    Knee.Replacement.Post.Op.Q.Limping +
    Knee.Replacement.Post.Op.Q.Work +
    Knee.Replacement.Post.Op.Q.Confidence +
    Knee.Replacement.Post.Op.Q.Shopping +
    Knee.Replacement.Post.Op.Q.Stairs +

```

```
Knee.Replacement.Post.Op.Q.Score) %>%
step_zv(all_predictors()) %>%
step_corr(all_predictors())
```

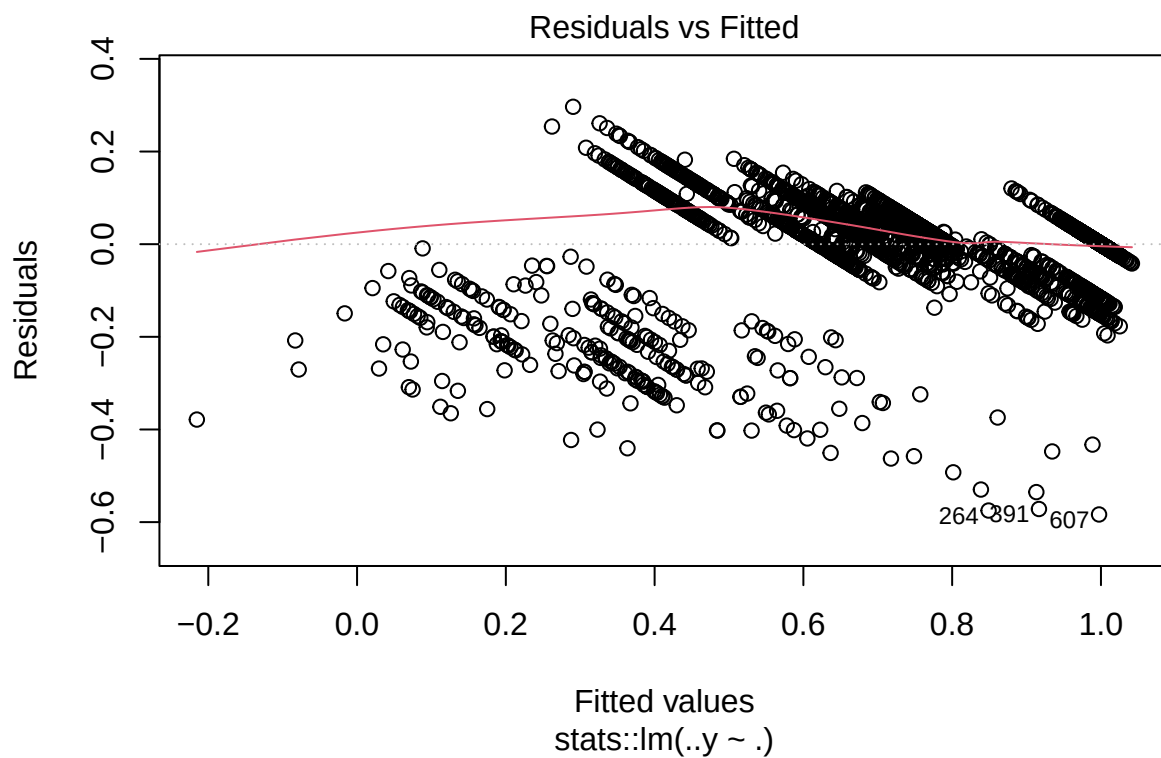
3. Model specification and training

```
linearModel <- linear_reg()

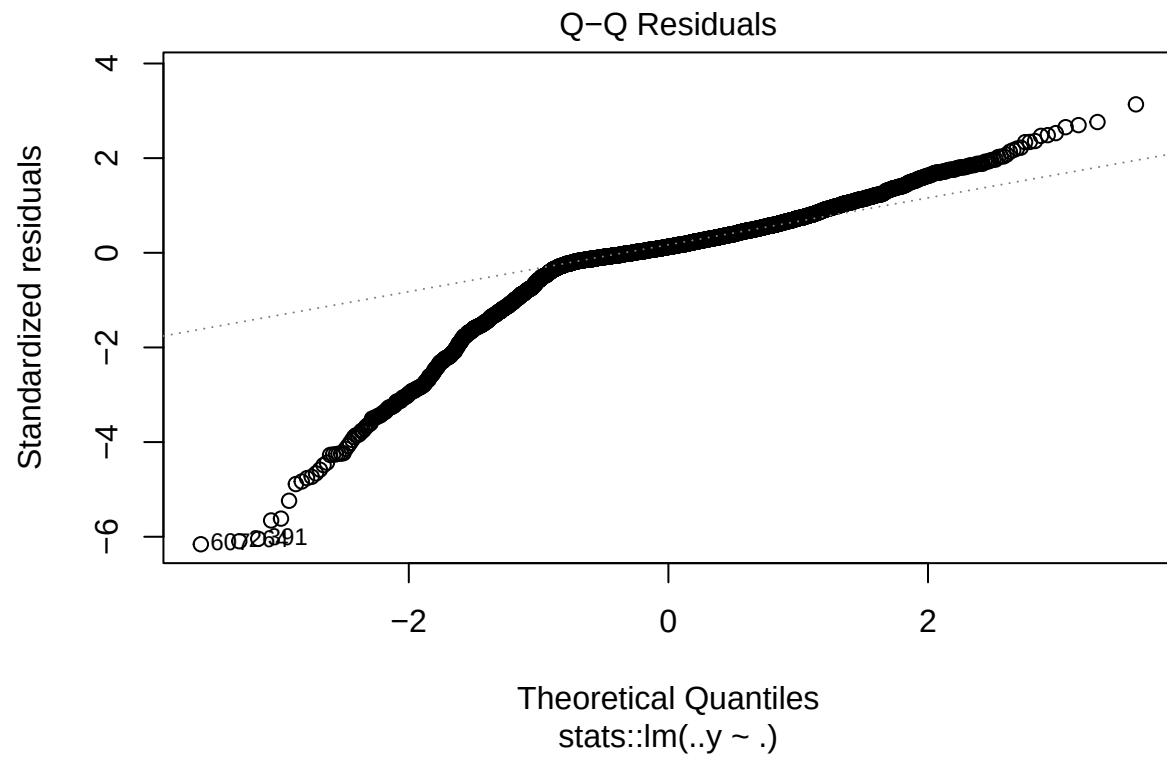
kneeLM <- workflow() %>%
  add_recipe(kneeModelRec) %>%
  add_model(linearModel)

kneeModelFit <- kneeLM %>%
  fit(data = kneeTrain)

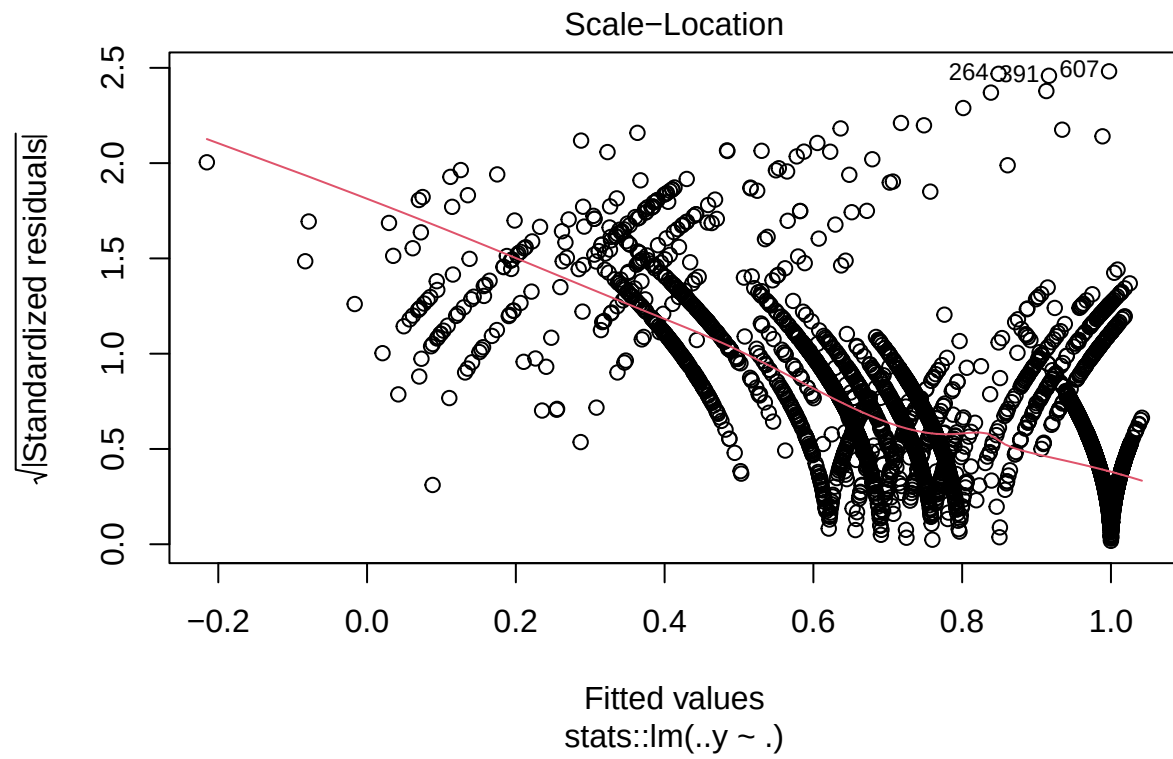
fittedKneeModel <- extract_fit_engine(kneeModelFit)
plot(fittedKneeModel, 1)
```



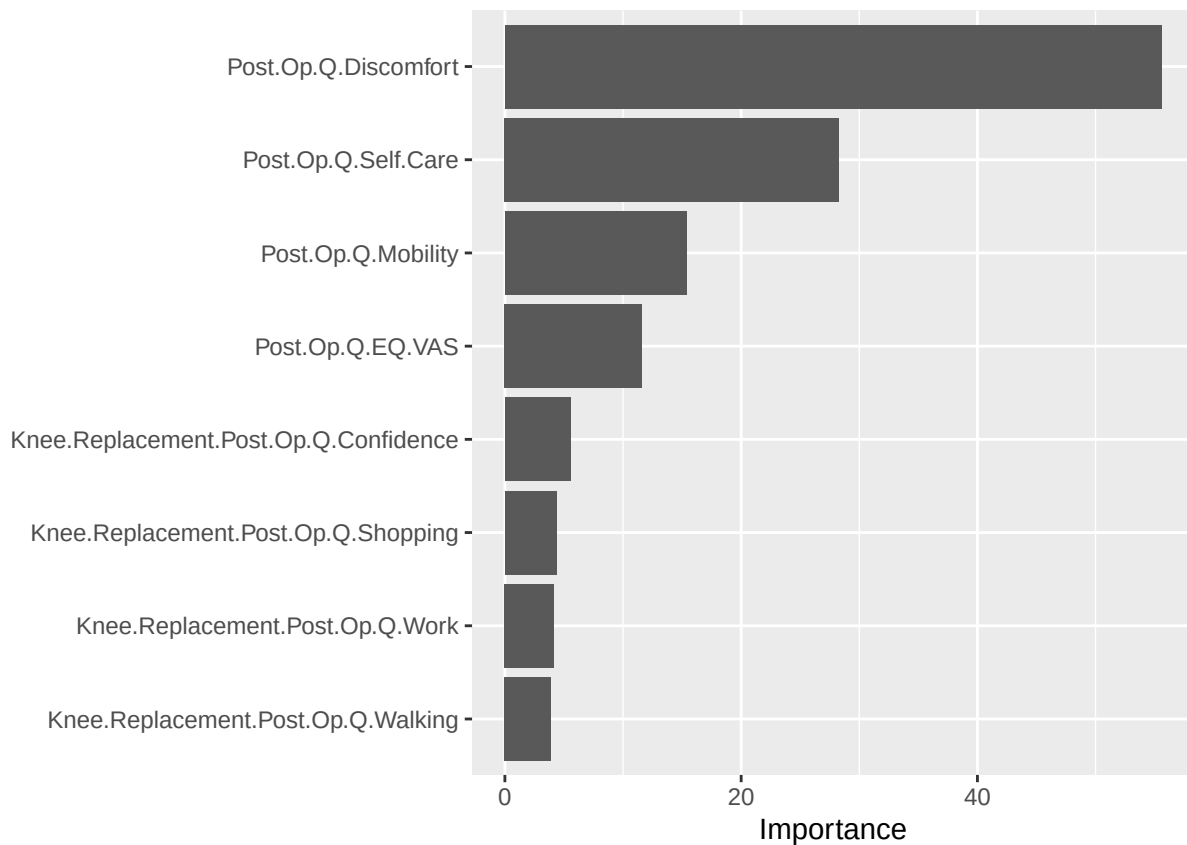
```
plot(fittedKneeModel, 2)
```



```
plot(fittedKneeModel, 3)
```



```
kneeModelFit %>% extract_fit_parsnip() %>% vip(num_features=8)
```



4. Model evaluation

```
predictedEQ <- kneeModelFit %>%
  predict(new_data = kneeTest)

resultsTest <- kneeTest %>%
  bind_cols(predictedEQ) %>%
  rename(kneeEQPred= .pred)

metrics(resultsTest, truth=Post.Op.Q.EQ5D.Index, estimate= kneeEQPred)
```

```
## # A tibble: 3 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 rmse    standard      0.104
## 2 rsq     standard      0.818
## 3 mae     standard      0.0668
```

Limitations of machine learning model